

PAPER_299 — Hydrogen Atom UQFF Electrogravitational Dominance Ratio: $\eta_{EM} = 9.65 \times 10^{29}$

Session: 85

Module: HYDROGEN_ATOM_UQFF_MODULE.cpp (27th C++ UQFF module — FIRST atomic-scale module)

System: Hydrogen ground state — Bohr model, $M_p = 1.6726 \times 10^{-27}$ kg, $r_{Bohr} = 5.2918 \times 10^{-11}$ m, $z = 0$

Category: Electrogravitational Dominance — FIRST UQFF atomic module, FIRST η_{EM} computation

UQFF Version: 2.0

Abstract

At the atomic scale, the UQFF framework reveals a fundamental asymmetry between gravitational and electromagnetic forces. The Newtonian gravitational acceleration at the Bohr radius ($g_{base} = 3.99 \times 10^{-17}$ m/s²) is the smallest base-gravity value across all 27 UQFF modules — five orders of magnitude below the previous minimum (M16 Eagle Nebula, 1.454×10^{-12} m/s²). The electron Lorentz acceleration in the atomic magnetic field ($a_{Lorentz} = q \times v_{orb} \times B / m_e = 3.85 \times 10^{13}$ m/s²) completely dominates the UQFF total. The ratio $\eta_{EM} = a_{Lorentz} / g_{base} = 9.65 \times 10^{29}$ defines the UQFF Electrogravitational Dominance Ratio — the largest force asymmetry computed in the UQFF framework.

1. Physical Setup

The hydrogen atom in its ground state is the smallest gravitational system tractable in the UQFF framework:

Parameter	Value	Units
Proton mass M_p	1.6726×10^{-27}	kg
Bohr radius r_{Bohr}	5.2918×10^{-11}	m
Electron mass m_e	9.1094×10^{-31}	kg
Electron orbital velocity $v_{orb} = \alpha \cdot c$	2.1877×10^6	m/s
Atomic magnetic field B_{atom} (est.)	1.0×10^{-4}	T
Electron charge q	1.6022×10^{-19}	C
Fine-structure constant α	7.2974×10^{-3}	—

2. Core Equations

2.1 Newtonian Base Gravity

$$g_{base} = \frac{G \cdot M_p}{r_{Bohr}^2} = \frac{6.6743 \times 10^{-11} \times 1.6726 \times 10^{-27}}{(5.2918 \times 10^{-11})^2}$$
$$g_{base} = \frac{1.116 \times 10^{-37}}{2.800 \times 10^{-21}} = 3.986 \times 10^{-17} \text{ m/s}^2$$

Smallest g_base in all UQFF modules. Prior minimum: M16 Eagle Nebula ($1.454 \times 10^{-12} \text{ m/s}^2$). Hydrogen is 5 orders of magnitude smaller.

2.2 Electron Lorentz Acceleration (Dominant Term)

$$a_{\text{Lorentz}} = \frac{q \cdot v_{\text{orb}} \cdot B}{m_e} = \frac{1.6022 \times 10^{-19} \times 2.1877 \times 10^6 \times 10^{-4}}{9.1094 \times 10^{-31}}$$

$$a_{\text{Lorentz}} = \frac{3.504 \times 10^{-17}}{9.109 \times 10^{-31}} = 3.848 \times 10^{13} \text{ m/s}^2$$

2.3 Electrogravitational Dominance Ratio [PAPER_299]

$$\eta_{\text{EM}} = \frac{a_{\text{Lorentz}}}{g_{\text{base}}} = \frac{3.848 \times 10^{13}}{3.986 \times 10^{-17}} = \mathbf{9.65 \times 10^{29}}$$

The electromagnetic Lorentz force exceeds Newtonian gravity by **30 orders of magnitude** at the hydrogen Bohr radius.

3. Computed Values

Quantity	Value	Units	Notes
g_base	3.986×10^{-17}	m/s ²	Newtonian at r_Bohr
a_Lorentz	3.848×10^{13}	m/s ²	DOMINANT EM term
η_{EM}	9.65×10^{29}	—	[PAPER_299]
a_Ug (Ug1+Ug4)	7.972×10^{-17}	m/s ²	2×g_base
a_quantum (HUP)	$\sim 1.5 \times 10^{-34}$	m/s ²	$(\hbar/\sqrt{(\Delta x \cdot \Delta p)}) \times 2\pi/t_H$
a_osc (Lyman)	$\sim 2 \times 10^{-10}$	m/s ²	[PAPER_300]
a_GR_min	$\sim 2.8 \times 10^{-60}$	m/s ²	[PAPER_301]
a_Lambda	3.30×10^{-36}	m/s ²	cosmological
Total g_H (t=0)	$\sim 3.85 \times 10^{13}$	m/s ²	EM-dominated

4. UQFF g_base Spectral Catalog

The hydrogen module defines the **EM-dominated regime** and the minimum g_base across all UQFF modules:

Module	System	g_base (m/s ²)	Session
Hydrogen (THIS)	H atom, r_Bohr	3.99×10^{-17}	85
M16 Eagle Nebula	nebula ~35 ly	1.454×10^{-12}	80
M87 (hypothetical)	galaxy	$\sim 10^{-10}$	—
HUDF z=3.5	deep field galaxy	$\sim 10^{-10}$	72g
Observable Universe	r= 4.4×10^{26} m	3.45×10^{-10}	84
Saturn	planetary	10.44	78

Hydrogen g_base is **5 orders smaller** than the prior minimum (M16) and **27 orders smaller** than Saturn.

5. Physical Interpretation

The η_{EM} ratio of 9.65×10^{29} quantifies the well-known dominance of electromagnetism over gravity at atomic scales. Within the UQFF framework, this is the first direct computation of this asymmetry as a UQFF term ratio. The result is consistent with the known force ratio (electromagnetic/gravitational $\approx 10^{36}$ for electron-proton, with η_{EM} here representing the Lorentz/Newtonian ratio specifically at v_{orb} and $B_{\text{atom}} = 10^{-4}$ T).

The EM term a_{Lorentz} represents the centripetal Lorentz force maintaining the electron in its Bohr orbit, expressed as acceleration per unit mass of the system (proton reference frame). This bridges classical orbital mechanics to the UQFF gravitational framework.

6. UQFF 2.0 Implementation

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// [PAPER_299] in HYDROGEN_ATOM_UQFF_MODULE.cpp updateCache():
g_base_cache      = G_NEWTON * M_proton / (r_Bohr * r_Bohr);      // 3.99e-17 m/s^2
a_Lorentz_cache   = Q_ELEM * v_orb * B_atom / m_elec;            // 3.85e13 m/s^2
eta_EM_cache      = a_Lorentz_cache / g_base_cache;              // 9.65e29 [PAPER_299]

WOLFRAM_TERM: HYDROGEN_LORENTZ = "a_Lorentz = q*v_orb*B/m_e = 3.85e13 m/s^2;
eta_EM = 9.65e29 [PAPER_299]"
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7. Significance

1. **First UQFF atomic-scale module:** Establishes the lower bound of the UQFF gravity spectrum
 2. **First η_{EM} computation:** Defines the electrogravitational dominance ratio as a named UQFF quantity
 3. **Minimum g_{base} :** $3.99 \times 10^{-17} \text{ m/s}^2$ — 5 orders below prior module minimum (M16)
 4. **EM-dominated limit:** All 26 prior modules were gravity-dominated or GR-dominated; hydrogen is the first EM-dominated
 5. **Scale anchor:** Provides the atomic-scale end of the UQFF multi-scale framework (companion to Universe-scale PAPER_296-298)
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8. Cross-References

- **PAPER_298** (Session 84): GR Curvature Dominance $\varepsilon_{\text{GR}} = 5.056 > 1$ at Universe scale — opposite extreme
 - **PAPER_301** (Session 85): Proton $\varepsilon_{\text{GR}} = 7.04 \times 10^{-44}$ — GR spectral minimum (same module)
 - **PAPER_288** (Session 81): DPM-THz Cascade, RSC module — prior smallest UQFF scale (~ 10 km plasma)
 - **PAPER_266** (Session 72g): Gravitational Meissner Effect — another EM-gravity interface
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9. Summary

$$\eta_{\text{EM}} = \frac{a_{\text{Lorentz}}}{g_{\text{base}}} = \frac{q \cdot v_{\text{orb}} \cdot B / m_e}{G \cdot M_p / r_{\text{Bohr}}^2} = 9.65 \times 10^{29}$$

The hydrogen atom UQFF module establishes the electrogravitational boundary: at the Bohr radius, the Lorentz electromagnetic force exceeds Newtonian gravity by 30 orders of magnitude. This defines the **EM-dominated regime** of the UQFF framework — the atomic-scale complement to the gravity-dominated and GR-dominated regimes of larger UQFF modules.

UQFF computed: UQFF energy correction term $[SSq]h?_g/(k_BT) = 0.57 \times 7.7e-50/(1.38e-23 \times 300) = 1.1e-29$; UQFF shift in Lyman-alpha = $1.1e-29 \times 13.6 \text{ eV}$.